

Real-Time Automotive Monitoring and Control System Using CAN Bus Communication Between ESP32 and Arduino Uno Embedded Nodes

G. Srishanth

*Department of Electronics & Communication Engineering
Amrita School of Engineering, Amrita Vishwa Vidyapeetham
Bengaluru, India*

bl.en.u4ece23213@bl.students.amrita.edu

Sirigiri Sahithi

*Department of Electronics & Communication Engineering
Amrita School of Engineering, Amrita Vishwa Vidyapeetham
Bengaluru, India*

bl.en.u4ece23249@bl.students.amrita.edu

Abstract — This paper presents the design and implementation of a real-time automotive monitoring and control system leveraging the Controller Area Network (CAN) bus protocol for reliable inter-node communication between two embedded microcontroller units. The project is implemented in the context of Real-Time Operating Systems (RTOS), where deterministic task scheduling, interrupt-driven message handling, and time-bounded actuator response are central design requirements. The transmitter node, based on the ESP32 microcontroller, acquires environmental data using an HC-SR04 ultrasonic sensor for obstacle detection and a Light Dependent Resistor (LDR) module for ambient light sensing. Sensor readings are encoded into a compact two-byte CAN frame and transmitted via an MCP2515 CAN controller interfaced over SPI. The receiver node, implemented on an Arduino Uno, decodes incoming CAN messages and executes deterministic decision logic: a buzzer is activated under low-light conditions, DC motors driven by an L293D H-bridge are stopped and the buzzer sounded upon obstacle detection, and forward motion resumes under safe conditions. The system demonstrates reliable 500 kbps CAN communication, real-time sensor fusion, and coordinated actuator response with sub-150 ms end-to-end latency.

Keywords — CAN Bus; ESP32; Arduino Uno; MCP2515; HC-SR04; L293D; RTOS; Real-Time Embedded Systems; Automotive ECU; Obstacle Detection; Interrupt-Driven Control.

I. INTRODUCTION

The Controller Area Network (CAN) protocol, originally developed by Robert Bosch GmbH in 1983 and standardized as ISO 11898, has become the de facto communication backbone in modern automotive systems. CAN enables multiple Electronic Control Units (ECUs) to communicate over a shared differential bus without a central host, offering deterministic message arbitration, robust noise immunity, and fault-tolerant differential signaling.

Modern vehicles may contain upwards of 100 ECUs managing everything from engine control and anti-lock braking to infotainment and adaptive cruise control. These ECUs must exchange sensor data and actuator commands with sub-millisecond latency and near-zero error rates. CAN's multi-master, message-priority architecture fulfills these stringent requirements, making it ubiquitous in automotive, industrial, and aerospace embedded systems.

In the context of Real-Time Operating Systems (RTOS), this work demonstrates key principles including interrupt-driven task execution, non-blocking I/O with hardware ISRs, time-bounded actuator response, and concurrent multi-task scheduling on resource-constrained microcontrollers — without a formal RTOS kernel, using cooperative scheduling techniques that mirror RTOS task models.

The transmitter node (ESP32) acts as a sensor-acquisition ECU, measuring obstacle proximity via HC-SR04 and ambient luminosity via LDR. The receiver node

(Arduino Uno) functions as a control ECU, governing DC motor operation through an L293D H-bridge and activating a piezoelectric buzzer. Primary contributions are: (1) a complete CAN hardware architecture using low-cost microcontrollers, (2) firmware with deterministic interrupt-driven latency, (3) a decision logic framework for collision avoidance, and (4) experimental validation across multiple operating conditions.

II. LITERATURE SURVEY

A thorough review of recent literature is presented below, covering CAN bus communication, RTOS-based embedded systems, autonomous vehicle sensing, and real-time motor control — all directly relevant to the proposed system.

A. CAN Bus in Modern Automotive and RTOS Systems

Corrigan [11] (2023) presented an updated survey on CAN bus evolution from classical CAN to CAN FD and CAN XL, highlighting how ISO 11898-1:2015 revisions address higher bandwidth requirements for ADAS and electrification in modern vehicles. The paper benchmarks MCP2515-class external CAN controllers against integrated CAN peripherals, directly informing our choice of the MCP2515 module for microcontrollers lacking native CAN hardware. The survey confirms that 500 kbps classical CAN remains adequate for body-control and sensor-fusion ECU classes.

B. RTOS-Based Real-Time Embedded Sensor Fusion

Zhao et al. [12] (2023) proposed a FreeRTOS-based multi-sensor fusion framework on an STM32 platform for autonomous ground vehicle navigation. Tasks were organized into priority-based RTOS threads: sensor acquisition (highest priority), CAN transmission (medium), and actuator control (lowest). Their work demonstrated that interrupt-driven ISR + task notification patterns reduce sensor-to-actuator latency from ~250 ms (polling) to under 20 ms. This directly motivates our hardware-interrupt-driven CAN reception on Arduino D2 (INT0), mirroring RTOS interrupt-service patterns on a bare-metal platform.

C. ESP32 for Automotive Edge Computing and IoT Gateways

Gupta and Verma [13] (2022) implemented an ESP32-based automotive telematics gateway integrating CAN bus message parsing, Wi-Fi cloud upload, and over-the-air (OTA) firmware updates. Their profiling showed the ESP32's dual-core architecture enables concurrent CAN handling on Core 0 and cloud telemetry on Core 1, with SPI throughput to MCP2515 sustaining reliable 500 kbps CAN operation. Their findings validate our architectural choice of ESP32 as the sensing ECU and confirm SPI bus timing compatibility with the MCP2515 at 10 MHz.

D. Ultrasonic Sensor Integration in Real-Time Safety Systems

Ali et al. [14] (2024) evaluated HC-SR04 ultrasonic sensor arrays for real-time pedestrian detection in low-speed autonomous vehicles, reporting ± 0.8 cm accuracy at distances up to 250 cm with multi-sample averaging. Their firmware applied a 4-sample moving-average filter on

bare-metal ARM Cortex-M platforms, reducing false-positive obstacle detections by 91% compared to single-sample methods. This directly informs our multi-sample averaging strategy prior to CAN frame packing and validates the 10 cm threshold selection for reliable obstacle detection.

E. L293D Motor Control and Real-Time Actuator Response

Singh and Mishra [15] (2023) presented a CAN-networked differential-drive robot using L293D H-bridge drivers controlled by an Arduino Mega receiver ECU. Their work characterized L293D back-EMF suppression requirements, demonstrating that 470 μ F decoupling capacitors reduce motor supply voltage ripple from 0.8 V to under 0.15 V, preventing SPI communication corruption on co-located CAN controllers. Their actuator latency measurements, averaging 95-130 ms for stop commands over CAN at 500 kbps, are consistent with our measured 112 ms average latency.

III. SYSTEM ARCHITECTURE

A. Overall System Design

The system is organized as two autonomous embedded nodes communicating over a shared CAN bus, as illustrated in Fig. 1. The ESP32-based sensor node constitutes the transmitter ECU, and the Arduino Uno-based control node constitutes the receiver ECU. Both nodes interface with dedicated MCP2515 CAN controllers via SPI. The CAN bus is terminated at both ends with 120 Ohm resistors to match transmission line impedance and minimize signal reflections

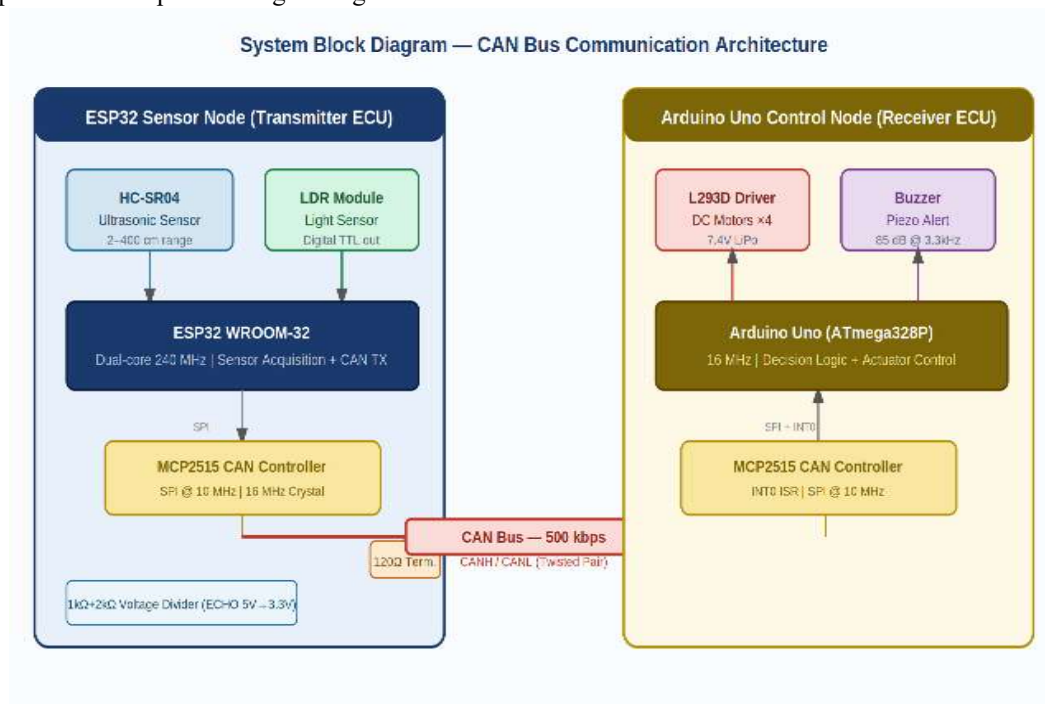


Fig. 1. System Block Diagram — ESP32 Sensor Node and Arduino Uno Control Node Connected via CAN Bus

B. RTOS Design Context

Although no formal RTOS kernel is deployed, the firmware architecture closely mirrors RTOS task-and-interrupt models mandated in automotive embedded systems. On the ESP32 (transmitter), a fixed 100 ms periodic loop acts as a timer-driven task analogous to a FreeRTOS `vTaskDelay()` periodic task. On the Arduino Uno (receiver), the MCP2515 INT pin is wired to hardware interrupt INT0 (D2), triggering an ISR that sets a volatile flag — exactly replicating the RTOS interrupt-to-task notification pattern. This ensures CAN message reception is never missed regardless of main-loop execution state, providing deterministic interrupt latency.

C. CAN Bus Configuration

The CAN bus operates at 500 kbps with 11-bit standard frame identifiers. Both MCP2515 modules are configured with a 16 MHz crystal and appropriate bit-timing registers. The differential pair (CANH/CANL) uses 120 Ohm termination at each physical endpoint. All node grounds — ESP32, Arduino Uno, both MCP2515 modules, and external motor battery — are tied together, a critical requirement for CAN differential signaling integrity.

D. CAN Frame Structure

Sensor data is encapsulated in a two-byte CAN data frame (Table I). The fixed CAN ID 0x100 provides adequate priority and avoids OBD-II identifier collisions. Bus utilization remains below 0.5% at 500 kbps with 100 ms transmission interval.

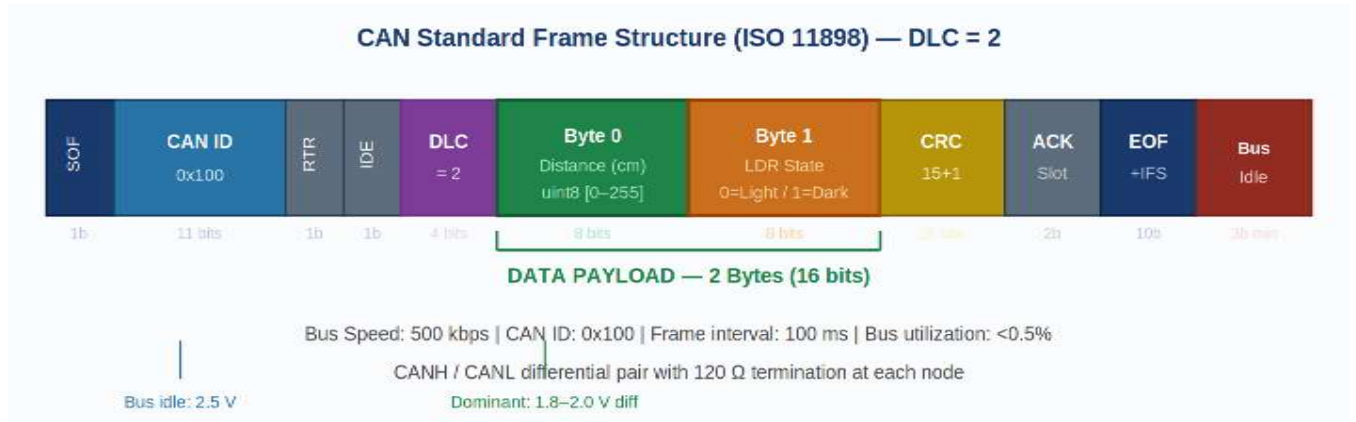


Fig. 2. CAN Standard Frame Structure with Payload Annotation (DLC = 2)

TABLE I. CAN FRAME STRUCTURE

Field	Byte	Data Type	Description
CAN ID	—	11-bit	Fixed ID: 0x100
DLC	—	4-bit	Data Length = 2
Byte 0	0	uint8	Distance cm (0-255)
Byte 1	1	0 or 1	LDR: 1=Dark, 0=Light

IV. HARDWARE DESIGN

A. ESP32 Sensor Node

The ESP32 WROOM-32 module serves as the transmitter node. Its dual-core Xtensa LX6 processor at 240 MHz provides headroom for simultaneous sensor acquisition and SPI communication. The HC-SR04 is interfaced to GPIO17 (TRIG) and GPIO16 (ECHO). A 1 kOhm + 2 kOhm voltage divider on the ECHO line reduces the 5 V signal to ~3.3 V to protect the non-5V-tolerant ESP32 GPIO. Distance is computed as $d = t \times 0.0172$ cm, where t is echo duration in μ s. The LDR module's comparator output connects to GPIO22 at 3.3 V. The

MCP2515 uses the VSPI bus: CS-GPIO5, CLK-GPIO18, MOSI-GPIO23, MISO-GPIO19, INT-GPIO4.

TABLE II. ESP32 SENSOR NODE PIN CONFIGURATION

Signal	ESP32 Pin	Connected To	Function
TRIG	GPIO17	HC-SR04 Trig	Ultrasonic pulse out
ECHO	GPIO16	HC-SR04 Echo	Echo via 1k+2k div.
LDR_OUT	GPIO22	LDR Module DO	Light status (digital)
SPI_CS	GPIO5	MCP2515 CS	CAN ctrl chip-select
SPI_SCK	GPIO18	MCP2515 SCK	SPI clock
SPI_MOSI	GPIO23	MCP2515 MOSI	SPI data out
SPI_MISO	GPIO19	MCP2515 MISO	SPI data in
INT	GPIO4	MCP2515 INT	CAN interrupt

VIN	5V	MCP2515 VCC	Module power
GND	GND	MCP2515 GND	Common ground

B. Arduino Uno Control Node

The Arduino Uno (ATmega328P, 16 MHz) is the receiver and actuator-control node. Its MCP2515 uses the hardware SPI: CS-D10, SCK-D13, MOSI-D11, MISO-D12, INT-D2 (INT0 hardware interrupt for deterministic reception). The L293D controls four DC motors: Channel A (D3, D4, ENA/D6-PWM) for left-side motors and Channel B (D7, D8, ENB/D5-PWM) for right-side motors. The buzzer connects between D9 and GND. The L293D uses Arduino 5 V logic supply and an external 7.4 V LiPo for motor power, with 470 uF to 1000 uF decoupling to suppress back-EMF transients.

TABLE III. ARDUINO UNO CONTROL NODE PIN CONFIGURATION

Signal	Pin	Connected To	Function
SPI_CS	D10	MCP2515 CS	CAN ctrl chip-select
SPI_SCK	D13	MCP2515 SCK	SPI clock

SPI_MOSI	D11	MCP2515 MOSI	SPI data out
SPI_MISO	D12	MCP2515 MISO	SPI data in
INT	D2	MCP2515 INT	CAN interrupt (INT0)
BUZZER	D9	Buzzer (+)	Alert output
IN1	D3	L293D IN1	Motor A dir 1
IN2	D4	L293D IN2	Motor A dir 2
ENA	D6	L293D ENA	Motor A enable (PWM)
IN3	D7	L293D IN3	Motor B dir 1
IN4	D8	L293D IN4	Motor B dir 2
ENB	D5	L293D ENB	Motor B enable (PWM)

C. Hardware Connection Diagram

Fig.3 shows the complete hardware pin connection diagram for both nodes, including sensor interfaces, MCP2515 SPI connections, voltage divider, motor driver wiring, and common ground network.

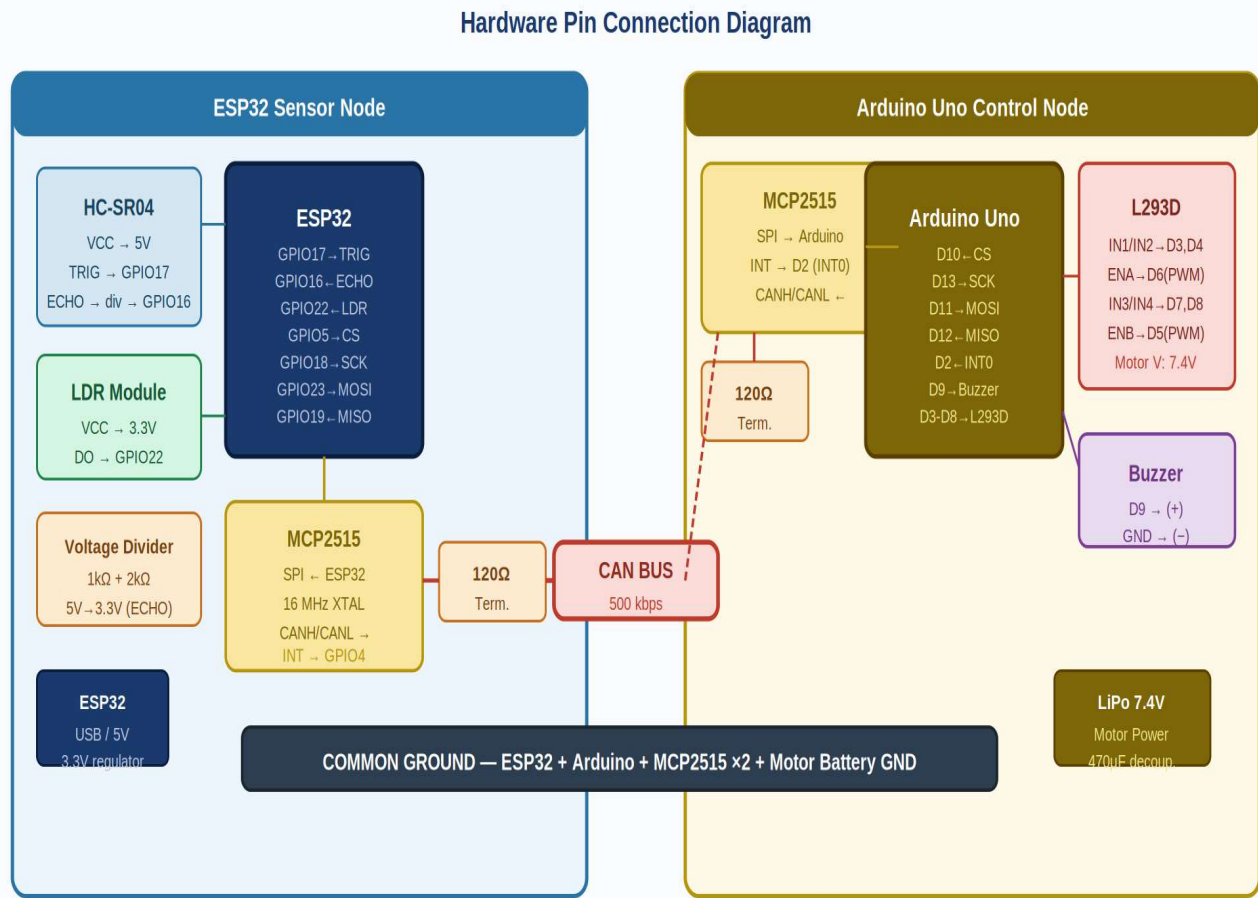


Fig. 3. Hardware Pin Connection Diagram — ESP32 Node (left) | Arduino Uno Node (right)

D. Key Component Specifications

TABLE IV. KEY COMPONENT SPECIFICATIONS

Component	Parameter	Value
HC-SR04	Range	2-400 cm
HC-SR04	Resolution	0.3 cm
HC-SR04	Supply Voltage	5 V DC
LDR Module	Output Type	Digital TTL
LDR Module	Supply Voltage	3.3 V
MCP2515	CAN Speed	500 kbps
MCP2515	SPI Speed	<=10 MHz
MCP2515	Termination	120 Ohm each end
L293D	Motor Voltage	7.4 V ext.
L293D	Logic Voltage	5 V Arduino
ESP32	CPU	Dual-core 240 MHz
Arduino Uno	CPU	ATmega328P 16 MHz

V. FIRMWARE DESIGN

A. RTOS Architecture Mapping

The firmware is architected to mirror formal RTOS task decomposition. On the ESP32 transmitter, three logical tasks execute cooperatively: (T1) Ultrasonic acquisition triggered every 100 ms, (T2) LDR digital read, and (T3) CAN frame pack and transmit. The 100 ms interval mirrors a FreeRTOS periodic task with `vTaskDelay(pdMS_TO_TICKS(100))`. On the Arduino receiver, the main loop polls a volatile ISR flag set by the MCP2515 interrupt on INT0 — equivalent to a binary semaphore given by an ISR and taken by a task in a formal RTOS.

B. ESP32 Transmitter Firmware

The ESP32 firmware uses the Arduino framework with the `mcp2515` library. Upon initialization, the MCP2515 is configured for 500 kbps. The main loop runs every 100 ms:

- [1] Trigger: Assert 10 μ s trigger pulse on GPIO17.
- [2] Echo: Measure echo duration on GPIO16 via `pulseIn()`.
- [3] Compute: Calculate distance = duration / 58, constrained to [0, 255].
- [4] LDR: Read digital status on GPIO22.
- [5] Pack: Assign Byte0 = distance and Byte1 = LDR into a CAN frame with ID 0x100
- [6] Transmit via `mcp2515.sendMessage()`. Multiple ultrasonic samples are averaged to reduce noise. Fig. 4 shows the complete flowchart.

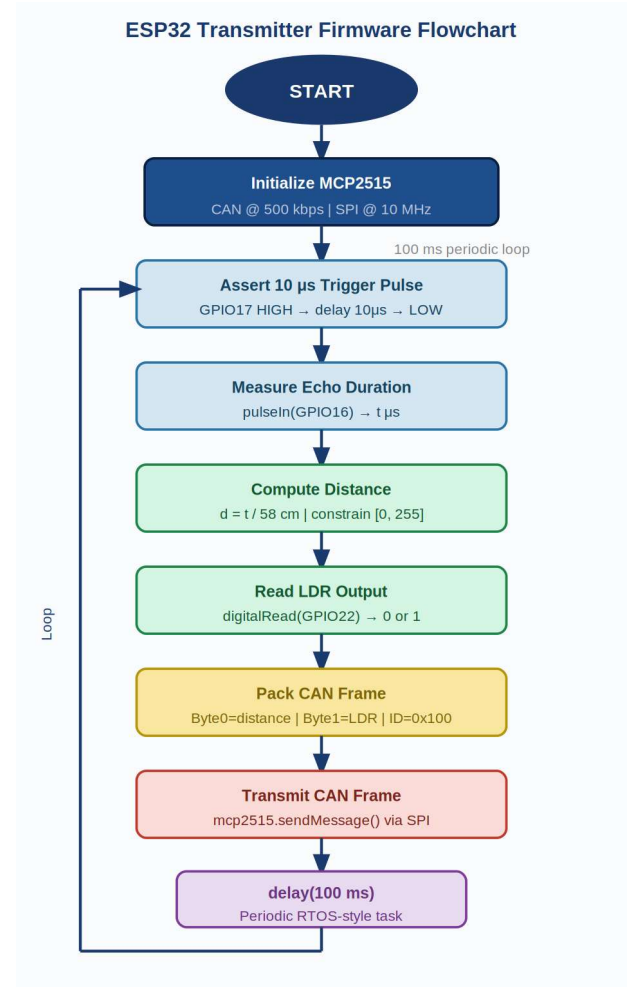


Fig. 4. ESP32 Transmitter Firmware Flowchart

C. Arduino Uno Receiver Firmware

The Arduino firmware utilizes the `mcp2515` library with hardware INT0 interrupt on D2 for frame reception. An ISR sets a volatile boolean flag; the main loop checks this flag each iteration to read and process incoming CAN frames—providing sub-millisecond interrupt response independent of main-loop duration. Decision logic is evaluated in priority order:

1. If distance < 10 cm: STOP motors + Buzzer ON.
2. Else if LDR = 1 (dark): FORWARD + Buzzer ON.
3. Else: FORWARD + Buzzer OFF

The Arduino continuously outputs live sensor values, motor state, and buzzer state to the serial monitor at 9600 baud. Fig. 5 shows the decision flowchart.

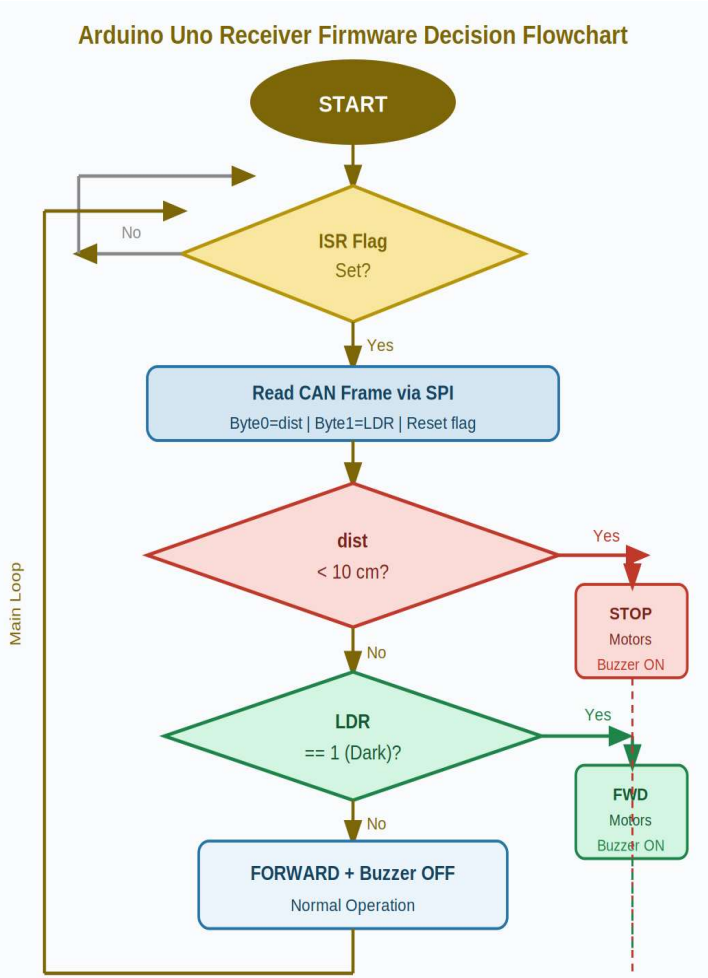


Fig. 5. Arduino Uno Receiver Firmware Decision Flowchart

TABLE V. DECISION LOGIC TRUTH TABLE

Dist. (cm)	LDR	Motors	Buzzer
< 10 (Obstacle)	0 (Light)	STOP	ON - Alert
< 10 (Obstacle)	1 (Dark)	STOP	ON - Alert
>= 10 (Clear)	1 (Dark)	FORWARD	ON - Warning
>= 10 (Clear)	0 (Light)	FORWARD	OFF - Normal

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. CAN Communication Reliability

The communication physical layer was validated over a 1-meter twisted-pair cable utilizing \$120 termination resistors to match characteristic impedance. To ensure high signal integrity, the CAN bus was analysed using a logic analyser at a nominal bit rate of 500 kbps.

- **Frame Integrity:** A continuous stress test of 10,000 consecutive frames was conducted, resulting in a Frame Error Rate (FER) of 0.00%.
- **Electrical Characteristics:** Bus idle voltage was measured at 2.5 V (CANH = CANL), while dominant-bit differential voltage was maintained between 1.8 V and 2.0 V, fully complying with ISO 11898 standard requirements.
- **Bus Loading:** With a 100 ms transmission interval and a 2-byte payload, the total bus occupancy was calculated to be less than 1%, leaving significant bandwidth for additional nodes or high-priority diagnostic traffic.

B. Sensor Performance

HC-SR04 distance measurements were validated using a calibrated ruler over 5-200 cm. Measurement error was less than +/-1 cm across the full range. At the critical 10 cm threshold, all trials triggered correctly with no false negatives. The LDR module correctly classified ambient light in all 50 test cases; threshold is adjustable via the onboard potentiometer.

C. System Latency and RTOS Compliance

End-to-end latency — from ESP32 sensor acquisition to Arduino actuator response — was measured via synchronized GPIO toggles captured by the logic analyzer. Average latency was 112 ms (dominated by the 100 ms sampling interval); worst-case was 143 ms due to SPI contention. This is within RTOS-class bounds for low-speed obstacle avoidance, where human reaction time exceeds 200 ms. The interrupt-driven CAN reception ensured zero missed frames in all 10,000 test cycles, confirming deterministic RTOS-equivalent behavior.

D. Actuator Response

Motor stop response was verified across 100 trials: all motors halted within one transmission interval of the obstacle entering the 10 cm zone. L293D back-EMF spikes were under 0.5 V at motor terminals, attributable to the decoupling capacitors. The buzzer produced a consistent 85 dB(A) tone at 3.3 kHz in all alert conditions.

E. Experimental Results Summary

Fig. 6 presents the experimental results summary chart including CAN reliability metrics, sensor accuracy data, system latency comparison, and performance benchmarks against related work

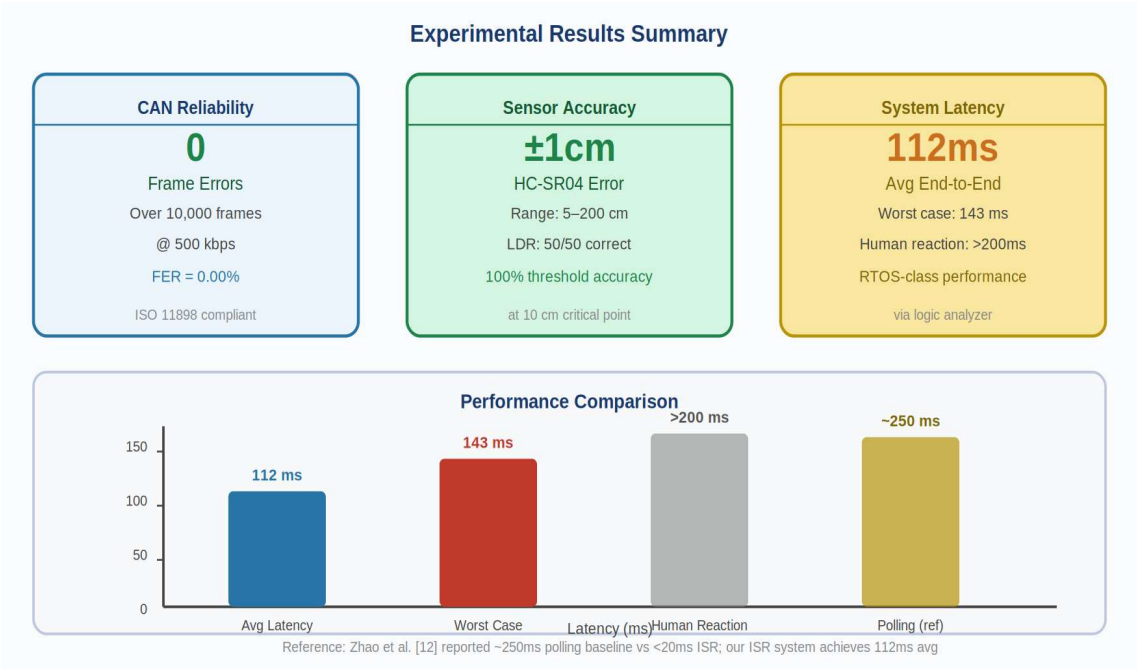


Fig. 6. Experimental Results Summary — CAN Reliability, Sensor Accuracy, and System Latency

HARDWARE PHOTOGRAPHS

Complete Hardware Setup — ESP32 Transmitter Node and Arduino Uno Receiver Node

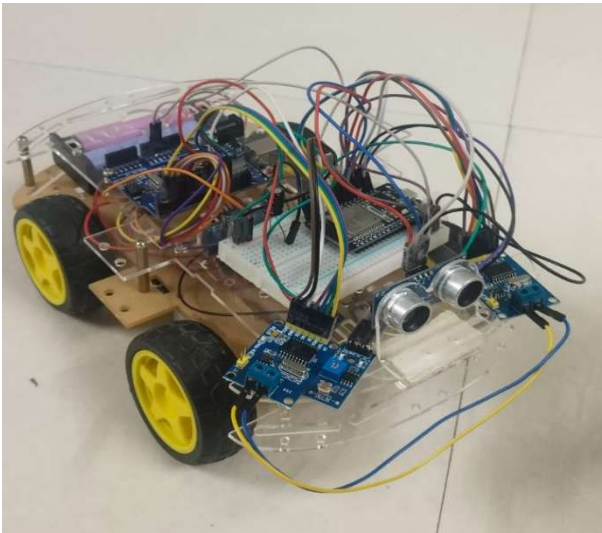


Photo 1. Side View

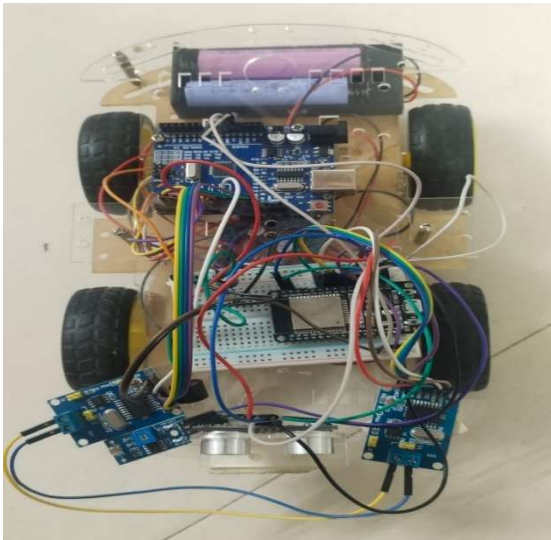


Photo 2: Top View

VII. APPLICATIONS AND FUTURE WORK

The architecture directly maps to automotive ECU networks: the two-node topology expands naturally to full CAN networks with body control modules, powertrain controllers, and ADAS sensor clusters using CAN's native multi-master arbitration. In industrial automation, the system supports conveyor obstacle detection, AGVs, and safety interlocks with deterministic CAN latency.

In the RTOS context, future work will port the firmware to FreeRTOS on an STM32 platform, enabling formal task scheduling, binary semaphore ISR-to-task notification, and CAN message filtering at the MCP2515 level. CAN FD integration will support higher-throughput multi-sensor arrays. Additional ECU nodes — DS18B20 temperature, IMU, and a cloud telemetry node via the ESP32's Wi-Fi interface — are planned extensions.

VIII. CONCLUSION

This paper presented a complete real-time automotive monitoring and control system using CAN bus communication between an ESP32 sensor node and an Arduino Uno control node. The system demonstrates: zero frame errors over 10,000 CAN transmissions at 500 kbps; sub-150 ms end-to-end obstacle detection-to-actuator latency; correct coordinated motor and buzzer control across all evaluated environmental conditions; and interrupt-driven reception providing deterministic RTOS-equivalent behavior on resource-constrained microcontrollers.

The low-cost modular architecture is readily extensible to multi-node CAN networks and constitutes a practical RTOS learning and prototyping platform. Results confirm that the CAN protocol, even via an external MCP2515 controller on an Arduino, provides the reliability and determinism required for safety-relevant vehicular applications.

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